

Cross Lingual Transfer in Tulu Legal Comprehension: Script-Dependent Improvement and RAG-Induced Knowledge Conflict

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Abstract

Low-resource languages without an adequate training corpus often use a related, higher-resource language as a scaffold for comprehension. Still, there is a need to develop rigorous evaluation methods to identify when models fail in cross lingual low-resource environments. Using the legal domain as a backdrop, three models (Llama3, Hex-1, Sarvam) were tested on the ability to classify legal complaints written in a low resource Dravidian language (Tulu). Transliterating queries across Dravidian scripts allowed models to gain a preliminary understanding of speakers' complaints without the use of wide scale training, though the level of comprehension was heavily script dependent (with Kannada - another relatively low-resource language - producing the strongest positive trend). Retrieving from a corpus of Kannada legal papers across a RAG framework caused mixed results. Some models had a weak positive trend in comprehension under certain conditions, but when models failed, it was often across two axes: fact substitution (fixating on specific passage excerpts that skewed reasoning) and confabulation (hallucination that had no basis in either query or corpus). Within low resource domains, results identify the model's parsing of information and subsequent reasoning as the source of reasoning failure, rather than corpus contents. Script-dependent comprehension and RAG robustness also seem to travel together. This is further supported by the reasoning-trace analysis and a statistical-honesty framework deployed - techniques that are more broadly applicable to low-resource multilingual RAG evaluation.

1 Introduction

Despite the various capabilities of large language models, one of the most pressing pitfalls is how models assess information in low-resource domains. In many regions of the world, the common language spoken (often a highly low-resource

language) differs from the administrative language. To exacerbate the issue, the administrative language is often low-resource itself, requiring novel techniques for models to bridge this cross-lingual low-resource domain. A relevant application of this dilemma is the legal field, where members of such communities often need to represent themselves in a non-native language to access resources.

This study uses the Dravidian language Tulu, a small language with a few million speakers off the coast of Karnataka. The official language of Karnataka is Kannada, meaning all legal proceedings take place with Kannada or English mediums. Kannada is itself classified a low resource language, as despite having an extensive literary history, training LLM on it has only picked up recently. Tulu is extremely low resource; it does not contain the extensive literature used to traditionally train models.

With a low resource language, Tulu, how would it be possible to file speakers' complaints into a system that does not administratively use their language (and instead uses Kannada), without a translator? In this study, we explore whether minimal priming and legal papers written in Kannada are enough for a model to accurately interpret queries written in a language they do not have previous exposure to.

While high-resource languages are often used as a starting point when working with low-resource languages, the intuition points in the other direction. When Tulu is written, it has traditionally used Kannada abugidas rather than the Latin alphabet. Tulu speakers are also more likely to borrow vocabulary and structure sentences similarly to the more common language around them - Kannada. Thus, we propose that use of the Kannada (and other Dravidian) scripts would trigger stronger comprehension in models. Furthermore, legal papers written in Kannada would provide a stronger foothold for an LLM to understand Tulu complaints spoken in

an informal register due to the aforementioned relation between the two languages.

The purpose of this paper is not to create a full, working LLM for low resource languages, but to evaluate model comprehension in a cross-lingual low-resource domain, using the legal field as an appropriate, grounded testing ground. To do so, we ask three LLM models to categorize legal complaints written in Tulu using the Legal Issues Taxonomy (LIST) from the Stanford Legal Design Lab (Hagan, 2026), without translation or heavy training. We explore whether minimal prompting and RAG from a corpus in a higher resource, related language, can serve as a scaffold for models to pinpoint the gaps in their understanding of the complaint.

Most Tulu speakers are multilingual and can communicate in Kannada, English, Hindi, etc, though doing so can still result in the loss of nuance that occurs when speaking in a second language.

2 Related Work

Knowledge Conflict - When training large language models, they often run into knowledge conflict, as shown by (Longpre et al., 2021). When models were forced to decide between a newly given passage or their own memory, they often fall back to their own understanding, even when prompted otherwise. However, later findings diverge from this understanding. (Xu et al., 2024) establishes the "context-memory conflict", where models struggle to properly weigh retrieved information that conflicts with their own internal understanding, and can inconsistently defer to context that is coherent and confidently stated, regardless of accuracy. This is further explored in (Wu et al., 2024), which found that models consistently abandon their correct prior knowledge in favor of retrieved context - including Llama3 - one of the models used in this study. (Xie et al., 2024) lends credence to the claim that models are receptive to confidently presented external evidence, even if a knowledge conflict is present, though notes another separate confirmation bias - models tend to stick with their own prior understanding when the evidence set only partially agrees with it. Given that model comprehension in a low resource, legal context is still not widely understood, our unique angle isolates how these mechanisms can be further utilized by comparing multiple models' innate ability, retrieval analysis, and discernment between the

two in a multilingual setting.

Effect of RAG Inclusion - RAG's inclusion can be detrimental to knowledge synthesis. (Hsia et al., 2025) explains how various model/retriever combinations can affect downstream performance, sometimes performing worse compared to no retrieval at all. Witnessed as well in (Zeng et al., 2025), where models performed worse when presented with misleading retrieved evidence compared to a zero shot baseline and could not recognize when to discard incorrect information. Our work furthers this discussion, displaying how models are able to use minimal priming to make their predictions more accurate, but the inclusion of RAG prevents most models from effectively discerning when to disregard irrelevant passages.

Hallucination in Legal NLP - It's especially vital to understand this issue in legal work, where small knowledge conflicts can snowball into major hallucinations. (Magesh et al., 2025) establishes how commercial legal AI built around RAG still meaningfully hallucinates, with literature like (Reuter et al., 2025) working towards mitigation. Our findings add to established literature, demonstrating that even relevant legal papers can cause hallucination within a model, which may be due to cross-lingual transfer or uncertainty in a low resource environment.

Retrieval and Low Resource Languages - This problem is magnified among low resource languages: (Sun et al., 2026) illustrate how low-resource language pairs rely more heavily on retrieved context and degrade more severely when that context is noisy compared to high-resource pairs. (Hu et al., 2025) extends this, demonstrating that while knowledge-free reasoning easily transfers across language despite the secondary resource effect, knowledge retrieval does not. This may explain why RAG frequently failed when the query was not written in a high resource language, which did not allow the model to discriminate between irrelevant passages and its own reasoning. Our methods also draw from (Qi et al., 2025), which separates context utilization from retrieval quality, though ours differ by maintaining consistent retrieval over queries written in multiple languages. Our findings are also supported by (Mohan et al., 2026), which explains how a failure occurs not from retrieved content itself, but how the model uses it - rationale for why introducing retrieved passages often failed, as the issue may lie with the model's ingestion of the passages as well

as confusion with multilingual transfer across a low resource domain. This intersection of low resource multilingual transfer across the legal domain is relatively unexplored.

Tulu-Kannada Relation - Within the scope of the Tulu-Kannada bridge, (Narayanan and Aepli, 2024) establishes that because of the extensive contact between the two languages, using Kannada as a scaffold for Tulu seems like a viable option. This is supported by (Devadiga and Chopra, 2026), which discovers how Tulu is "nearly indistinguishable" from Kannada to Kannada trained LLM. Our results align, given the drop in comprehension when Tulu is written in other Dravidian scripts compared to Kannada.

3 Methodology

3.1 Task Formulation

To discover to what extent a model could understand queries in Tulu, we authored 60 sentences describing common legal situations, with each sentence ascribed a primary category, inspired by the Legal Issues Taxonomy (LIST) from the Stanford Legal Design Lab (Hagan, 2026) (where some categories were replaced), as well as any relevant secondary categories. The model had to use the sentence given (as well as any additional context) to predict what category the Tulu complaint fell under, with performance evaluated across the 60 sentences. The hope was to build a pipeline that could classify complaints in a low resource language with minimal prompting.

3.2 Data and Models

The 60 Tulu sentences were written in an informal, common register describing a situation requiring legal analysis. The sentences were written in Latin script as well as three Indic scripts, primarily Kannada, but also Malayalam and Tamil, and translated into English as a control.

The legal corpus consisted of 18 legal documents from Karnataka courts (taken from ecourtsindia.com), written in Kannada. Documents were orders of various sizes, ranging from a paragraph and a table describing common proceedings to 25-page hearings detailing witness testimonies. Many corpus documents consisted of images or used legacy non-Unicode fonts that produced mojibake when extracted (garbled Latin text rather than standard Kannada), requiring automated text extraction. To clean up extraction

based errors, AI assisted correction was utilized, an established technique (Kanerva et al., 2025). Using Gemini, original documents were given alongside the scraped text for the model to identify and correct text based errors, ensuring the model had a reference to source material rather than open-ended inference.

The cleaned legal papers were each ascribed to at least one major LIST category, although secondary categories could be ascribed as well. The documents exist solely for the model to gain an understanding of similar problems to the query, to aid diagnosis.

Three models were used to understand cross-lingual legal transfer. Llama3 (AI@Meta, 2024) had no official support for Kannada or Tulu, though a small portion (over 5%) of its training data covered more than 30 additional languages not officially documented. Sarvam (Sarvam AI, 2024) and Hex-1 (Bud Ecosystem, 2025) were both trained with a focus on Indic languages including Kannada, but not Tulu, testing whether a Kannada basis helps comprehension. All three models had temperature 0.1 for reproducibility; Llama3 and Hex-1 had a repeat penalty of 1.1. Hex-1's context window (num_ctx) was raised to 8192 to prevent truncation. Enabling thinking mode on Hex-1 caused inner reasoning with no response, while Sarvam's reasoning consumed all tokens in its "starter tier" (4000 tokens), also yielding no output, so thinking/reasoning was disabled for both.

For embedding the legal texts, we originally used nomic-embed-text, but switched to multilingual-e5-large-instruct, as the latter produced embeddings spanning a wider range rather than deferring to a default median.

3.3 Experimental Conditions

Each condition was composed of four parts: (1) the script and language the query was written in, (2) whether the model received linguistic priming, (3) whether the model received RAG passages from the Kannada legal corpus, and (4) if primed, whether the priming was sparse or extensive.

Priming had two variants. **Sparse** priming contained a short paragraph of common Tulu vocabulary (most not present in the test sentences) and basic grammatical structure. **Extensive** priming was a continuation of sparse priming that additionally included words related to describing legal situations - e.g. family, land, permission - some of which were used directly in the 60 test sentences.

Tulu words in the priming block matched the query’s script (e.g. a Romanized-Tulu query received Romanized-Tulu priming vocabulary), while instructions remained in English throughout. The English control never received priming.

Retrieved passages shown to the model included the source case name, retrieval similarity score, and the passage’s own LIST category label. Baselines included English-gloss queries with RAG (Condition A), to test how RAG affected comprehension with a high-resource query language, and Romanized/Kannada-script queries without priming or RAG (Condition G and Condition H, respectively), to test latent understanding without modification.

Condition	Script	Priming	RAG
A	English gloss	–	✓
B	Romanized Tulu	–	✓
C	Romanized Tulu	✓	✓
D	Kannada script	✓	✓
E	Kannada script	✓	–
F	Kannada script	–	✓
G	Romanized Tulu	–	–
H	Kannada script	–	–
I	Tamil script	–	–
J	Malayalam script	–	–
K	Tamil script	✓	–
L	Malayalam script	✓	–
M	Romanized Tulu	✓	–

Table 1: Experimental conditions, defined by query script, presence of linguistic priming, and use of retrieval-augmented generation (RAG). Conditions with priming (C, D, E, K, L, M) were each run at both sparse and extensive tiers.

3.3.1 Prompting

Each model was first given the query, after which retrieved RAG passages were provided if applicable. If RAG was used, the model was instructed not to classify the passages, but to use them to understand the situation described in the query. The model was again given the query, then presented a third time with explicit instructions not to classify retrieved passages. This repeated instruction was necessary, as otherwise the model often resolved to classify its interpretation of retrieved passages.

3.4 Retrieval Pipeline

All legal passages were first broken into 800 character chunks, which were embedded using nomic-embed-text, then switched to multilingual-e5-large-instruct (the query was embedded similarly). The embedded query was compared against

chunks and the top 3 chunks were retrieved using cosine similarity. Mean centering was briefly deployed to test whether it increased the number of relevant passages retrieved, but provided no recognizable optimization.

3.5 Diagnostic Methodology

To assess whether the model was able to comprehend the query, its response was analyzed across four separate axes.

Tertiary Classification - Since queries could often relate to multiple categories, each query was given two other related LIST labels, alongside its primary LIST classification. The model, after being provided the query and possible subsequent material, had to provide a primary, secondary, and tertiary LIST category, and could not repeat categories. If the primary category provided by the model matched the assigned primary classification, it received a point to its Hits@1 score. If the model’s primary classification did not match, but the subsequent guesses corresponded with any of the three categories assigned to the query, it received a point to the Hits@3 score.

A high Hits@1 and Hits@3 score indicated the model was able to successfully assign the proper LIST category, as well as any tangentially related categories. A high Hits@3 but low Hits@1 score was seen as the model partially understanding the query, but not enough to capture the nuance described. A low Hits@1 and Hits@3 score meant the model did not have strong comprehension of the queries.

Reasoning Trace Verification - Models were instructed to provide reasoning for their primary guess, to ensure whether the model had actually understood the situation, or produced an unrelated reason that coincidentally aligned with the query. A model with a high number of such situations was seen as partially parsing the query, but then hallucinating the gaps in its comprehension.

Statistical Honesty - To ensure that discrepancies between conditions were significant and not due to random error, Fisher’s Exact Test was deployed (e.g. comparing Condition E and Condition D for a given model). Binomial tests were also used, comparing a model’s D-condition performance against random guessing (12.5% accuracy) and majority class guessing (25% accuracy, assigning the most common category across queries). Multiple runs for a specific model/condition pair were pooled for stronger statistical analysis. The p

value must be less than the standard threshold of 0.05 to be statistically significant. Findings that qualified as significant were described with confident language (e.g. "demonstrates", "shows"), while those that did not were viewed as directional or indicative of a trend, even if raw percentage gaps seemed large.

3.6 Preliminary Mitigation

After the discovery of RAG hurting a model’s discernment, a tentative prototype was tested.

CRAG-Style Relevance Verification - A secondary judge layer was added before the prompt, in which the model discerned whether each retrieved passage was relevant with a binary judgment (for 20 complaints only). Each call had two forms: one required the model to use queries written in Kannada script, the other used the English (Condition A) queries. Each passage was reviewed via independent calls. Non-relevant passages were excluded, and if all passages were deemed irrelevant, the prompt fell back to the No-RAG format. This was tested on the Llama3 D conditions (RAG inclusive & priming).

4 Findings

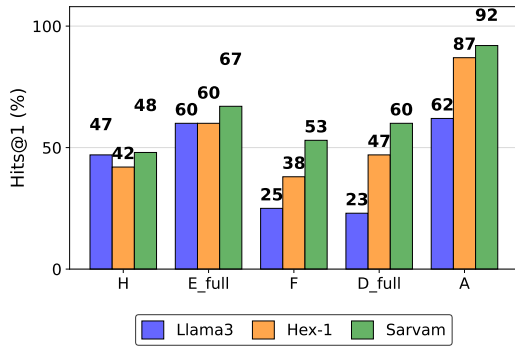


Figure 1: Hits@1 across five conditions per model ($n = 60$ each): bare comprehension (H), no-RAG full-priming ceiling (E_full), RAG without priming (F), RAG with full priming (D_full), English-gloss query with RAG (A). Only Llama3 drops below its own no-RAG ceiling once RAG is added. Full 19-condition breakdown: Appendix D.

Across all three models, Kannada-script bare comprehension (Condition H) is significantly above both chance and majority-class baselines (Llama3 46.7%, Hex-1 41.7%, Sarvam 48.3%; $p < .001$ in every case), and full-tier priming reveals a trend with comprehension being increased further for every model, when using Kannada (Figure 1).

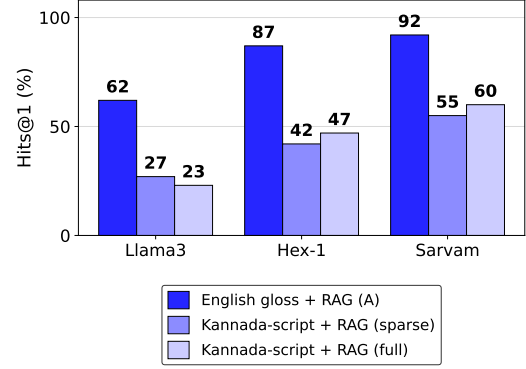


Figure 2: Query language under RAG: English-gloss queries (Condition A) substantially outperform Kannada-script Tulu queries under identical retrieval (Condition D), for all three models (Fisher’s exact, $p < .001$ in every comparison). This is the one RAG-related effect that generalizes across models.

The picture changes sharply once RAG is introduced. For Llama3, RAG collapses performance well below its own no-RAG ceiling (D_full 23.3% vs. E_full 60.0%, $p < .001$); Hex-1 and Sarvam show no significant degradation on the same comparison ($p = .20$ and $p = .57$, respectively). The full 19-condition, four-script, three-model breakdown and complete significance tests are reported in Appendix D.

The one RAG-related effect that generalizes across all three models is query language: English-gloss queries under RAG (Condition A) significantly outperform Kannada-script Tulu queries under otherwise identical retrieval (Condition D/Full), for every model ($p < .001$ in all three comparisons; Figure 2). A full comparison of Kannada and Romanized script performance across all priming and RAG conditions is reported in Table 10. This suggests that whatever degrades performance under RAG interacts more consistently with query language than with model identity.

5 Discussion

5.1 Comprehension Transfers Without RAG

One pressing finding was the repeatable success of Condition E. This trend emerges with Condition H, where models were able to classify given sentences without any scaffolding (at 46.7%, 41.7%, and 48.3% for Llama3, Hex-1, and Sarvam respectively). In other cases, the model was able to partially understand the sentence, but not enough to accurately diagnose it. To ensure this was not random chance, we also checked the reasoning given.

Query (gloss)	Reasoning trace
"I don't know which court I should go to"	"The situation explicitly mentions 'court', indicating a problem related to the legal system."
"Someone is living on my land without asking and won't leave"	"...classic housing/trespass issue."

Table 2: Sarvam, Condition H, ids 14 and 17. Row 1: human chosen and predicted category both Courts/Legal-System, reasoning names the specific recognized term rather than a generic paraphrase. Row 2: human chosen - Land/Property, predicted - Housing. The model parses the core action but assigns an adjacent category, indicating partial rather than absent comprehension.

Still, this was a preliminary indication that models were somehow able to occasionally parse queries correctly, despite no priming or previous training with the language.

Once the model received priming (Condition E), Llama3's and Hex-1's understanding and subsequent diagnosis improved slightly (46.7%→51.7% and 41.7%→45.0%, respectively, though neither reaches significance: $p = .715$ and $p = .854$), and this is not reflected with Sarvam (48.3%→46.7%, $p = 1$).

We had two theories as to how the model understood Tulu with brief exposure to the language's structure and no previous interactions with complete sentences in the target language.

The first proposed that the model was parsing Tulu as a form of Kannada, since the shared history between the two languages facilitated heavy word borrowing.¹ The second theory asserts that the model is reading Tulu as a unique Dravidian language, using features shared across the Dravidian family to make sense of the language. Given that Tulu exists in the same language family as Tamil and Malayalam (as well as Kannada), this also seemed viable.

Given the results of Condition I-L, script matters, though not substantially at the sparse-priming tier, and this holds across all three models. Sparse structural priming, grammar and word order only with no category-revealing vocabulary, produced only modest, non-significant changes in Kannada accuracy for every model (Llama3 46.7%→51.7%, $p = .715$; Hex-1 41.7%→45.0%,

¹See (Devadiga and Chopra, 2026), who find Kannada-script Tulu "nearly indistinguishable from Kannada" to models trained predominantly on the latter.

$p = .854$; Sarvam 48.3%→46.7%, $p = 1$), and left Tamil (Llama3 16.7%→15.0%; Hex-1 28.3%→33.3%; Sarvam 28.3%→33.3%; all $p > .26$) and Malayalam (Llama3 21.7%→20.0%; Hex-1 25.0%→28.3%; Sarvam 21.7%→38.3%; all $p > .07$) similarly flat or only weakly positive. If priming were teaching models portable Tulu grammar, this improvement should not be script-dependent. Since the difference does not reach significance for any script or model, this is only suggestive of script-specific assimilation over broad cross-Dravidian transfer. This effect was comparable with Romanized Tulu (Llama3 16.7%→13.3%, $p = 1$; Hex-1 16.7%→18.3%, $p = 1$; Sarvam 33.3%→30.0%, $p = .845$).

It is undeniable that models do have latent capability for understanding queries in this low-resource language, although the workings are still unclear. Parsing through a related, higher-resource language that the target language is lexically (rather than grammatically) similar to, and specifically through matching script, seems likely, although the relatively strong comprehension among other Dravidian scripts cannot be disputed.

5.2 RAG Causes a Large, Reproducible Collapse or No Significant Improvement

Llama3's comprehension dropped when presented with RAG, illustrated by comparing Condition D to F (F 25.0% vs. D_sparse 26.7%, $p = 1$; vs. D_full 23.3%, $p = 1$), where priming did not help under RAG as it did in Condition E (H 46.7%→E_full 60.0%), though this did not reach significance. The issue also arises in the control (Condition A): even when the query is written in a high-resource language, a comparatively lower-resource retrieval corpus still degraded the model's own understanding of the query: Llama3 reaches only 61.7% in this condition, well short of Hex-1's 86.7% and Sarvam's 91.7% under the identical retrieval setup. Meanwhile, both Sarvam and Hex-1 were able to maintain similar scores with the inclusion of RAG compared to just priming (Hex-1: F 38.3% vs. E_sparse 45.0%, $p = .579$), and priming seemed to hurt Sarvam (Conditions E_sparse vs. F: 46.7% vs. 53.3%, $p = .584$), though not significantly. This may be due to Sarvam and Hex-1 being trained on Kannada, allowing stronger discernment between passages and their own reasoning, which may also explain Llama3's heavier drop in Condition A.

A disagreement-level diagnostic further con-

518 firms this, which determined the ratio of Con- 569
519 dition E (non-RAG/priming) vs Condition D 570
520 (RAG/priming) producing the right answer when 571
521 disagreement occurred. D failed more often E 572
522 on almost all comparisons when disagreement oc- 573
523 curred - as high as 6.5:1 with Llama3 at full prim- 574
524 ing - except Sarvam and sparse priming in which 575
525 this trend reverses (0.4:1), with the full Table 11 576
526 reflecting model robustness under RAG. 577

527 When failure due to RAG did occur, the idea 578
528 that the cause was knowledge conflict, not retrieval 579
529 noise, is supported across several dimensions. 580

530 A model’s garbled post-RAG responses fell into 581
531 two categories: fact substitution and confabula- 582
532 tion. With **fact substitution**, the model takes a 583
533 specific but wrong detail present in the retrieved 584
534 passage and incorporates it directly into its reason- 585
535 ing, treating the snippet as part of the query rather 586
536 than supporting evidence. With **confabulation**, 587
537 the model hallucinates information present in nei- 588
538 ther the query nor the retrieved passage, using it as 589
539 a basis for reasoning regardless. This occurs even 590
540 when a genuinely relevant passage is retrieved: for 591
541 two debt-related queries, the retrieved case (*Srid- 592*
542 har P. v. Ramesh, see Appendix E) was a real, on- 593
543 topic cheque-bounce dispute, yet the model still 594
544 produced unrelated categories for both. 595

545 Methodological note: retrieved passages were 596
546 shown to the model together with their own source 597
547 category label. A follow-up ablation stripping this 598
548 label from the retrieved context produced no con- 599
549 sistent change in accuracy (deltas within $\pm 5pp$ 600
550 across all five RAG conditions, no consistent direc- 601
551 tion), indicating label-copying was not the primary 602
552 driver of the collapse; the failure is substantially 603
553 content-level, not a result of label exposure. 604

554 5.3 Relation of RAG Inclusion and Script 605 555 Across Model Performance 606

556 While the independent effects of RAG inclusion 607
557 and script have been stated, the intersection be- 608
558 tween the two highlights a possible relation - pri- 609
559 marily evident between Kannada script and Latin 610
560 script queries. With non-RAG conditions, Kan- 611
561 nada script significantly outperformed Latin script 612
562 across majority of model and condition combina- 613
563 tions Table 10, lending more credence to the dis- 614
564 cussion present in §5.1. Models can and do exhibit 615
565 comprehension of low resource languages that they 616
566 had previously never encountered, but the inclu- 617
567 sion of Latin script adds a complicating factor - that 618
568 models struggle to parse queries in a low-resource 619
620

language if the script the language is written in is 569
far removed from related languages in the model’s 570
training base (though comprehension can still ex- 571
ist). 572

573 However, this difference in comprehension be- 574
575 tween script is not universally present with RAG 576
577 inclusion. Sarvam and Hex-1 - both models that up- 578
579 held performance with RAG inclusion (with com- 580
581 parison between D and E conditions)- were also 582
583 models that could maintain the Kannada script ad- 584
585 vantage with RAG conditions (across Conditions 585
586 C and D in Table 10). Llama3 - the model in 587
588 which accuracy decreased (Condition D vs E), was 589
589 also the only model that could not significantly 590
590 beat the Latin query performance present in Con- 591
591 dition C (though the raw numbers of Condition D 592
592 were still greater). Whether a model is able to suc- 593
594 cessfully overcome knowledge conflict and utilize 594
595 the script advantage seem to be correlated. Fur- 595
596 thermore, whether a model is able to uphold per- 596
597 formance with RAG inclusion is also an indicator 597
598 of whether script advantage will remain under the 598
599 same RAG conditions. 599

599 It should be noted that while Llama3’s perfor- 600
600 mance dropped - resulting in insignificance - Sar- 601
601 vam’s performance was remarkably high in Latin 602
602 script queries (once again causing insignificance). 603
603 So while an insignificant increase in three condi- 604
604 tions exists with Sarvam (Condition E/(no RAG & 605
605 sparse priming), Condition H (no RAG & no prim- 606
606 ing), and Condition D/(RAG & full priming), the 607
607 underlying cause differed from Llama3. It’s pos- 608
608 sible that Sarvam’s explicit training in Kannada 609
609 allowed it to understand some of the Latin script 610
610 queries. 611

611 Why and when a model is able to understand 612
612 a script with non-RAG conditions but not under 613
613 RAG conditions is still unclear. Both Sarvam 614
614 and Hex-1 are trained in Kannada, which may al- 615
615 low them to maintain script advantage even along- 616
616 side complex legal passages. Since parsing Tulu 617
617 queries as Kannada produces more accurate re- 618
618 sults, the ability to uphold the script advantage 619
619 could also cause the model to overcome the knowl- 620
620 edge conflict present in RAG conditions, hence 621
621 why neither model was significantly affected by the 622
622 inclusion of RAG. Llama3 - the one model not ex- 623
623 plicitly trained in Kannada - may not have a strong 624
624 enough basis to recognize the query language and 625
625 corpus language to be similar even with the same 626
626 script, hence the drop in accuracy. However, this is 627
627 entirely speculative, and more work must be done 628

to understand this phenomenon.

5.4 Mitigation of RAG Failure

Using Corrective Retrieval-Augmented Generation (CRAG), models tested with Kannada-script queries performed similarly to the unmitigated RAG condition (25.0% Hits@1, vs. 23.3% for D/full). There was an improvement (to 60.0%) when the judge operated on English-translated complaints instead of Kannada-script queries, matching Llama3’s own no-RAG ceiling (E/full, 60.0%), even though RAG was still used in 17 of 20 cases; this difference does not clear significance at the standard threshold ($p = .054$, Fisher’s exact), though it sits at the boundary. The English-gloss judge also discerned more passages to be irrelevant on average (1.65 of 3 retained, vs. 2.85 of 3 for the Kannada judge with full results in Table 12), suggesting the judge layer’s ability to discriminate relevant from irrelevant retrieved content depends on the language it operates in, not solely on the underlying passage relevance.

6 Conclusion

In communities where the spoken language differs from the administrative language, training legal models on the latter does not address the issue of communities needing to use these tools in a cross lingual manner in courts, testimonies, etc. Common languages spoken in the affected area (Tulu, in this case) may not even have a literary history, making large scale training of models difficult. With the language pair of Kannada (administrative) and Tulu (spoken), the two share a long history of word borrowing and a similar syntactical structure. Considering this basis, is it possible to bridge the gap between two low resource languages, and alleviate the difficulty of switching between vernacular and formal legal speak with RAG and minimal priming alone? The strategy seems viable, although it heavily depends on the model used.

Given the task of classifying Tulu queries written in an informal register into specific LIST categories, models could be given two forms of priming, RAG retrieved passages, both priming and the inclusion of RAG, or neither technique at all. Surprisingly, all three models (Llama-3, Hex-1, and Sarvam) displayed relatively high levels of comprehension of the queries even with no priming or RAG at all, and both extensive and sparse priming produced an improvement over bare compre-

hension within conditions utilizing Kannada script for most conditions and models, though this effect did not reach significance for any model in Kannada script. However, RAG had the opposite effect on Llama3. Despite preliminary evidence that Tulu is parsed as Kannada by LLMs, the inclusion of actual Kannada legal papers only served to confuse the model on two fronts: fact substitution and confabulation. This may be because the model, treating Tulu as a form of Kannada, overly trusted retrieved passages over its own reasoning. Meanwhile, Sarvam and Hex-1 were not negatively or positively affected by RAG inclusion, though the same two mechanisms still appeared. Why Sarvam and Hex-1 did not produce the same RAG induced collapse as Llama3 is still unclear, though comparison of performance between Romanized and Kannada script queries signals that latent comprehension between scripts and the effect of RAG inclusion are interrelated rather than independent phenomena. While Llama-3 did exhibit comprehension with Kannada script, this effect did not significantly manifest once RAG was included. Hex-1 was able to maintain the advantage of Kannada script under both RAG and non-RAG conditions as well as Sarvam at most tiers, beating performance under Romanized queries. It is possible that due to training on Kannada, both Sarvam and Hex-1 were able to use internal knowledge and treat external sources as confirmation rather than a definitive answer. Further work must be done to understand this trend.

Llama3 (the only model hallucination reduction attempts were studied on) was somewhat responsive to the CRAG-style mitigation approach. CRAG Relevance Verification indicated the model could more easily discern irrelevant passages when the query was written in English, though this effect did not transfer to Kannada script.

Research in multi-lingual legal transfer that does not require heavy training is necessary, as affected communities often do not have the means to do so. In future work, we hope to continue bridging this gap, exploring cross-Dravidian transfer, larger datasets, more extensive RAG hallucination mitigation methods, and different language pairs to isolate the mechanisms and make such techniques generalizable to the greatest number of communities affected by this issue.

Limitations

Due to the time intensive need to carefully review queries to ensure native-like production, accurate grammar, and correct transliteration, it was considered unrealistic to generate an extensive number of queries for a pilot study. This study originally employed only 20 queries to test comprehension, but this number was deemed to be too small to be able to confidently assess results. Therefore, the final number of 60 queries was used to draw preliminary conclusions, although many of the findings were indicative of trends rather than statistically significant behavior, and would require a number of queries that enters the hundreds or thousands range to be considered significant. However, for large effects, such as the drop in Llama3 comprehension in condition D, these reach statistical significance even with a small dataset. Even so, if this study was to be expanded, having a larger number of queries would be a top priority.

The hallucination findings of this study do raise the question of whether similar risk was present during corpus creation, as final documents were scanned but not thoroughly reviewed by a native speaker. Kanerva et al. (Kanerva et al., 2025) similarly found LLM-based OCR correction to be highly effective for a well-supported language (English) while not reaching such levels in a lower-resource language (Finnish). It should be noted that Gemini officially supports Kannada (Deccan Herald, 2024). Additionally, the task required character verification rather than semantic deduction, distinguishing this corpus-creation context from that failure case. We still note this does not fully eliminate the residual risk discussed above and maintaining accuracy in future studies requiring a similar corpus should be a priority.

As for the legal documents used themselves, their public availability raises some concerns. Specifically, there is no way to verify if the models used in this study had been previously trained on said documents. Performance under RAG-inclusive conditions could be confounded if models were previously exposed to documents present in the corpus, and the non-RAG/RAG divide discussed in this paper should be revisited if this is found to be the case.

With regards to models, Sarvam and Hex-1 had their reasoning/thinking modes disabled, as compared to Llama3 which was allowed unconstrained free-form generation. Therefore, Sarvam's and

Hex-1's responses should be analyzed knowing that they were not produced under equivalent conditions.

Ethics Statement

While all 60 queries are authored to represent realistic hypothetical scenarios, they are not real individuals' complaints and do not draw from real case intake data. We believe this is the best way to maintain accurate analysis without the need for private individuals' legal situations. Meanwhile, corpus documents were directly retrieved from ecourtsindia.com, where court orders are publicly accessible. Retrieved passages were not anonymized in this study. Court orders were only used to extract document content and LIST category relevant information, and personal information was never used as a basis for classification. We acknowledge this as a limitation and recommend that named-entity redaction be utilized before using retrieved passages in further studies.

It should be heavily noted that this work is not deployment ready, given our findings of model hallucination present throughout experimentation. Deploying such pipelines has high chances of directing clients to LIST categories irrelevant to their query. Furthermore, due to the identified phenomena of fact substitution and confabulation, models can confidently provide incorrect reasoning, which can further mislead clients who do not have extensive experience in the legal field.

While prompt construction and priming methodology are detailed in Appendix A.2 and Appendix B, all scripts used during experimentation will be accessible in their entirety upon publication, as well as the 60 authored sentences. As for the legal corpus, the redistribution rights of the papers remain unclear, although case identifiers for examples discussed in Appendix E are available upon request.

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A Full Query Set and Priming Glossary

A.1 The 60-Sentence Query Set

Table 3 lists all 60 Tulu legal-complaint sentences used in this study, with their human chosen LIST primary and secondary categories, English gloss, and Romanized-Tulu form. Table 4 gives the corresponding Kannada script form.

Table 3: Full 60-sentence query set: human chosen LIST categories, English gloss, and Romanized Tulu.

ID	Primary	Secondary	English Gloss	Romanized Tulu
1	Housing	Consumer/ Money/ Debt	My landlord did not give back my money	enna dhani kasn vapas kortijer.
2	Work/ Employment	–	My employer did not pay me this month	enna dhani i tingolda sanbalan kortijer.
3	Torts/ Individuals	–	Someone broke my fence and did not fix it	era enna belin hal maltd nanla sari maltijer.
4	Consumer/ Money/ Debt	–	I gave someone money and they disappeared	yan erega kas kori bokka akulu tappayer.
5	Land/ Property	–	My neighbor took my land	enna baritakulu enna jagan detter.
6	Torts/ Individuals	–	After someone hit me with a stick my hand hurts	era enk badutu aki bokka enna kai bene avondundu.
7	Work/ Employment	–	My boss told me to leave without any reason	enna dhani karana korande yan kelasa budodu pand pander.
8	Consumer/ Money/ Debt	–	The person I lent money to is not returning it	yan sala kori bokka enk kas vapas korondijjer.
9	Housing	–	Someone came into my house without my permission	enna oppige ijjande ero enna illda ulayi batter.
10	Consumer/ Money/ Debt	–	They want more money than what we decided	akaleg enkulu nirnaya maltina aidd jasti kas kenonduler.
11	Consumer/ Money/ Debt	Courts/ Legal- System	They ended our agreement before the time we decided	akulu enkulu nirnaya maltina samayadda dunbe oppandan mugipayer.
12	Courts/ Legal- System	Consumer/ Money/ Debt	I signed a paper but did not understand what it said	yan daskat padye, matra kakajid dada baretd pand gottiji.
13	Consumer/ Money/ Debt	–	After I signed the paper they did not do what they agreed to	yan kakajid daskat padi bokka enkulu tirpu maltina vishayan akulu malpere kenondijer.
14	Courts/ Legal- System	–	I don't know which court I should go to	enk vovu nyayalayag podu pand gottiji.
15	Courts/ Legal- System	Consumer/ Money/ Debt	The person who owes me money is in a different place	enadda sala detondina jana bete jagad uller.

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Table 3 – continued

ID	Primary	Secondary	English Gloss	Romanized Tulu
16	Torts/ Individuals	Housing	My neighbor's tree fell on my house and damaged it	enna baritaklena mara enna illda mitt burda enna illan hal maltnd.
17	Land/ Property	Housing	Someone is living on my land without asking and won't leave	ero enna oppige ijjande enna jagad kulonda, itte budda popujer.
18	IP/Identity	–	Someone is selling their things using my name	ero aklena samann enna puarda maronduler.
19	Work/ Employment	–	My old employer is stopping me from starting my own work	enna paratta dhani enna svanta vyaparan suru malpere budondijer.
20	Estates/ Wills	Land/ Property	After my father passed away my family took the land that was meant to be given to me	enna popa tir poyi bokka enk barpina jagan enna kutunbadakulu detonyer.
21	Housing	Consumer/ Money/ Debt	My landlord is trying to raise my rent before our agreement ends	enklena tirpu maltina samayada dunbu enna dhani badige jasti malpere prayatna maltonduller.
22	Housing	Torts/ Individuals	The people next door make so much noise I cannot sleep at night	baritakulu et bobbe padonduller panda enk ratred jeppere avondiji.
23	Housing	–	My landlord will not fix the broken pipe in my kitchen	enna adipilda hal ayina nallin enna dhani sari malpere tayar ijer.
24	Housing	–	I was told to leave my rented house with only two days notice	enk badigeda illan budiye khali radd dinata vayide koryer.
25	Housing	Consumer/ Money/ Debt	My roommate moved out and left me to pay all the rent	enna sahavasi ill buddu badadigen pura enna mitte padd poyer.
26	Consumer/ Money/ Debt	–	The shop owner sold me a broken phone and won't give my money back	enk onji hal ayna phon-n mard daangadida dhani kasn pira korere tayar ijer.
27	Consumer/ Money/ Debt	–	I paid for a service that was never completed	yan pura kas kori bokkala belen purti maltijer.
28	Consumer/ Money/ Debt	–	My brother borrowed money from me two years ago and still has not paid	enna megge radd varshada dunbu ennadda sara dettonde, aven nanla pira kortije.
29	Consumer/ Money/ Debt	–	The bank is charging me fees I never agreed to	enna oppandina shulkag byank enna mitt kas padondundu.
30	Consumer/ Money/ Debt	–	Someone convinced me to invest in something that turned out to be fake	ero enan onji dongi astida mitt kas padere oppayer.
31	Work/ Employment	–	My workplace has not paid me for the extra hours I worked	yan maltina jasti kelsag enk beleda jagadd kas tikkiji.
32	Work/ Employment	–	I was fired the same week I told my boss I was sick	yan dhanita yan hushar iiji pandd pandina varade ar enan beledd detter.

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Table 3 – continued

ID	Primary	Secondary	English Gloss	Romanized Tulu
33	Work/ Employment	–	My employer is making me work without the safety equipment I was promised	enna dhani enan suraksha dipina saman korper pand pandi bokka aven korande enada bele malpavonduller.
34	Work/ Employment	–	I was promised a promotion but someone else got it without explanation	enk onji promoshan korper pandd patera koryer, matra dala karana korande bete janak koryer.
35	Work/ Employment	Consumer/ Money/ Debt	My employer refuses to give me my final paycheck after I quit	yan belen budi bokka enna dhani enna akherida sanbalan korondijjer.
36	Land/ Property	–	My neighbor built a wall that crosses onto my land	enna nerekaledakulu aklana belin enna jagada ullayi mutta kattader.
37	Land/ Property	–	Someone is tilling on land that belongs to my family without asking	enda kenande yera enna kutunbada jagad benni dattonduller.
38	Land/ Property	–	The boundary markers on my property were moved without my knowledge	enna jagada belin yera dett budter.
39	Land/ Property	Estates/ Wills	My brother is claiming a part of our family's land is his alone	enna palaye enna kutunbada jagada onji bhagan ayena matra pandd panondulle.
40	Land/ Property	Torts/ Individuals	Someone cut down trees on my property without permission	enda kenande yera enna jagada maran kadter.
41	Torts/ Individuals	–	My neighbor's dog bit my child while we were walking	enkulu nadetonduttinapaga enna baritaklena nayi enna balen cuccind.
42	Torts/ Individuals	–	Someone crashed into my parked vehicle and drove away	ero enklena untundu ittina karg akd poyer.
43	Torts/ Individuals	–	A worker dropped something that injured me	onji beledaye enna mitt saman budpad enk tagnd.
44	Torts/ Individuals	–	Someone spread lies about me that damaged my reputation in the village	hallid yera enna bagge sullu pandd enna pudarn hal malter.
45	Torts/ Individuals	Land/ Property	My neighbor's cattle keep destroying my crops	enna nerekaleda aklana pettalulu enna pura bulen tindd hal maltnd.
46	Courts/ Legal- System	–	I received a court notice but I don't understand what it is asking me to do	enk nyayalayada onji kakaji baidnd, matra ait dada panonduller pandd gottiji.
47	Courts/ Legal- System	–	I missed a court date because no one explained when I needed to appear	enk nyayalayag popina tarikh tatnd, dayeg panda enk erela epa e podu pandd pandijjer.
48	Courts/ Legal- System	Work/ Employment	I don't know how to file a complaint against my employer	enk enna kelsada dhanina mitt enca duru korodu pandd gottiji.

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Table 3 – continued

ID	Primary	Secondary	English Gloss	Romanized Tulu
49	Courts/ Legal- System	–	The police would not register my complaint	polis-da akulu enna durun dettonere tayar ijjer.
50	Courts/ Legal- System	–	I was told I need a lawyer but I don't know how to find one	enk onji wakil bodu pandd pander, matra enk vakiln enca nadodu pandd gottiji.
51	IP/Identity	Consumer/ Money/ Debt	Someone opened a loan in my name without my knowledge	enk gottijande yera enna pudard sala dettonyer.
52	IP/Identity	–	My photo is being used in an advertisement without my permission	enada kenande yera enna photon jahiratud upayoga maltonduller.
53	IP/Identity	–	Someone is pretending to be me online to trick my relatives	yera ennaleka anlain-d natane maltd enna kutunbadaklen dongi maltonduller.
54	IP/Identity	–	A shop is using my family's business name without asking	vova onji angadi enna kutunbada angadida pudaran upayoga maltondu.
55	IP/Identity	–	Someone stole the design of the products my family makes and sells	enklena kutunba maltd maruna samanda disain-n yera kandder.
56	Estates/ Wills	–	My father died without a will and my siblings are dividing everything without me	enna popa onji vil budande tir poyi bokka enna meggepalayenakulu enan buddu astin pura pattonduller.
57	Estates/ Wills	–	My mother's jewelry was promised to me but my brother took it after she died	enna ammana bangar enk korper pand pandi bokka amma tir poyi bokka enna palaye aven pura dettonye.
58	Estates/ Wills	Land/ Property	I found out my parents' house was sold after my father's death without our agreement	enna popa tir poyi bokka enna oppige ijande enna popa ammana illan mayer pandd gottand.
59	Estates/ Wills	Land/ Property	My uncle is refusing to share the property left by my grandfather	enna mamu, ajjer-na jagan pattere kenondijjer.
60	Estates/ Wills	Housing	After my husband passed away his family is not letting me stay in our house	enna kandani tir poyi bokka enna kandanina kutunbadakulu enan enklena illad kulyere budondijjer.

Table 4: Full 60-sentence query set in native script (Kannada).

ID	Kannada
1	ಎನ್ನ ಧನಿ ಕಾನ್ಸ್ ವಾಪಸ್ ಕೊರ್ತೀಜೀರ್.
2	ಎನ್ನ ಧನಿ ಈ ತಿಂಗೋಲ್ದ ಸಂಬಳನ್ ಕೊರ್ತೀಜೀರ್.

continued on next page

Table 4 – continued

ID	Kannada
3	ಏರಾ ಎನ್ನ ಬೇಲಿನ್ ಹಾಳ್ ಮಲ್ತ್ ನನ್ನ ಸರಿ ಮಲ್ತಿಜೇರ್.
4	ಯಾನ್ ಏರೇಗಾ ಕಾಸ್ ಕೊರಿ ಬೊಕ್ಕ ಅಕುಲು ತಪ್ಪಾಯೇರ್.
5	ಎನ್ನ ಬರಿತಕುಲು ಎನ್ನ ಜಾಗನ್ ದತ್ತೇರ್.
6	ಏರಾ ಎಂಕ್ ಬಡುಟು ಆಕಿ ಬೊಕ್ಕ ಎನ್ನ ಕೈ ಬೇನೆ ಆವೊಂದುಂಡು.
7	ಎನ್ನ ಧನಿ ಕಾರಣ ಕೊರಂದೆ ಯಾನ್ ಕೆಲಸ ಬುಡೊಡು ಪಂದ್ ಪಂಡೇರ್.
8	ಯಾನ್ ಸಾಲ ಕೊರಿ ಬೊಕ್ಕ ಎಂಕ್ ಕಾಸ್ ವಾಪಾಸ್ ಕೊರೊಂದಿಜ್ಜೇರ್.
9	ಎನ್ನ ಓಪ್ಪಿಗೆ ಇಜ್ಜಂದೆ ಏರೋ ಎನ್ನ ಇಲ್ಲ ಉಲಯಿ ಬತ್ತೇರ್.
10	ಅಕಲೆಗ್ ಎಂಕುಲು ನಿರ್ಣಯ ಮಲ್ತಿನ ಐಡ್ ಜಾಸ್ತಿ ಕಾಸ್ ಕೇನೊಂದುಲೇರ್.
11	ಅಕುಲು ಎಂಕುಲು ನಿರ್ಣಯ ಮಲ್ತಿನ ಸಮಯಡ್ ದುಂಬೇ ಓಪ್ಪಂದನ್ ಮುಗಿಪಾಯೇರ್.
12	ಯಾನ್ ದಸ್ತಕ್ ಪಾಡ್ಕೇ, ಮಾತ್ರ ಕಾಕಜಿಡ್ ದಾದಾ ಬರೆತಂಡ್ ಪಂದ್ ಗೊತ್ತಿಜಿ.
13	ಯಾನ್ ಕಾಕಜಿಡ್ ದಸ್ತಕ್ ಪಾಡಿ ಬೊಕ್ಕ ಎಂಕುಲು ತಿರ್ಪು ಮಲ್ತಿನ ವಿಷಯನ್ ಅಕುಲು ಮಲ್ತೇರೆ ಕೇನೊಂದಿಜೇರ್.
14	ಎಂಕ್ ಪೋವೊ ನ್ಯಾಯಾಲಯಗ್ ಪೋಡು ಪಂದ್ ಗೊತ್ತಿಜಿ.
15	ಎನಡ್ ಸಾಲ ದೆತೊಂದಿನ ಜನ ಬೇತೆ ಜಾಗಡ್ ಉಲ್ಲೇರ್.
16	ಎನ್ನ ಬರಿತಕ್ಕೇನ ಮರ ಎನ್ನ ಇಲ್ಲ ಮಿತ್ ಬೂರ್ಡ್ ಎನ್ನ ಇಲ್ಲನ್ ಹಾಳ್ ಮಲ್ತಂಡ್.
17	ಏರೋ ಎನ್ನ ಓಪ್ಪಿಗೆ ಇಜ್ಜಂದೆ ಎನ್ನ ಜಾಗಡ್ ಕುಲೊಂದು, ಇತ್ತೇ ಬುಡ್ ಪೋಪುಜೇರ್.
18	ಏರೋ ಅಕ್ಕೇನ ಸಾಮಾನ್ ಎನ್ನ ಪುರಡ್ ಮಾರೊಂದುಲೇರ್.
19	ಎನ್ನ ಪರಾತ್ತ ಧನಿ ಎನ್ನ ಸ್ವಂತ ವ್ಯಾಪಾರನ್ ಸುರು ಮಲ್ತೇರೆ ಬುಡೊಂದಿಜೇರ್.
20	ಎನ್ನ ಪೋಪ ತೀರ್ ಪೋಯಿ ಬೊಕ್ಕ ಎಂಕ್ ಬರ್ಪಿನ ಜಾಗನ್ ಎನ್ನ ಕುಟುಂಬದಕುಲು ದೇತೋಣ್ಣೇರ್.
21	ಎಂಕ್ಲೇನ ತಿರ್ಪು ಮಲ್ತಿನ ಸಮಯದ ದುಂಬು ಎನ್ನ ಧನಿ ಬಾಡಿಗಿ ಜಾಸ್ತಿ ಮಲ್ತೇರೆ ಪ್ರಯತ್ನ ಮಲ್ತೊಂದುಲೇರ್.
22	ಬರಿತಕುಲು ಏತ್ ಬೊಬ್ಬಿ ಪಾಡೊಂದುಲೇರ್ ಪನ್ನ ಎಂಕ್ ರಾತ್ರೆಡ್ ಜಿಪ್ಪೇರೇ ಆವೊಂದಿಜಿ.
23	ಎನ್ನ ಅಡಿಪಿಲ್ಲಾ ಹಾಳ್ ಆಯಿನ ನಳ್ಳಿನ್ ಎನ್ನ ಧನಿ ಸರಿ ಮಲ್ತೇರೆ ತಯಾರ್ ಇಜ್ಜೇರ್.
24	ಎಂಕ್ ಬಾಡಿಗಿದ ಇಲ್ಲನ್ ಬುಡಿಯೇರೆ ಖಾಲಿ ರಡ್ ದಿನತ ವಯಿದೇ ಕೊರ್ಯೇರ್.
25	ಎನ್ನ ಸಹವಾಸಿ ಇಲ್ಲ ಬುಡ್ ಬಾದಡಿಗಿನ್ ಪುರಾ ಎನ್ನ ಮಿತ್ತೆ ಪಾಡ್ ಪೋಯೇರ್.
26	ಎಂಕ್ ಒಂಜಿ ಹಾಳ್ ಆಯ್ ಪೋನ್-ನ್ ಮಾರ್ದ್ ದಲಂಗಡಿದ ಧನಿ ಕಾಸ್ ಪಿರಾ ಕೊರೆರೆ ತಯಾರ್ ಇಜ್ಜೇರ್.
27	ಯಾನ್ ಪುರಾ ಕಾಸ್ ಕೊರಿ ಬೊಕ್ಕಲಾ ಬೇಲಿನ್ ಪೂರ್ತಿ ಮಲ್ತಿಜೇರ್.
28	ಎನ್ನ ಮೆಗ್ಗೆ ರಡ್ ವರ್ಷದ ದುಂಬು ಎನ್ನಡ್ ಸಾರ ದೆತ್ತೊಂಡೆ, ಅವೆನ್ ನನ್ನಾ ಪಿರಾ ಕೊರ್ತಿಜಿ.
29	ಎನ್ನ ಒಪ್ಪಂದಿನ ಶುಲ್ಕಗ್ ಬ್ಯಾಂಕ್ ಎನ್ನ ಮಿತ್ ಕಾಸ್ ಪಾಡೊಂದುಂಡು.
30	ಏರೊ ಎನನ್ ಒಂಜಿ ಡೋಂಗಿ ಆಸ್ತಿದ ಮಿತ್ ಕಾಸ್ ಪಾಡೆರೆ ಒಪ್ಪಾಯೇರ್.
31	ಯಾನ್ ಮಲ್ತಿನ ಜಾಸ್ತಿ ಕೆಲ್ಸಗ್ ಎಂಕ್ ಬೇಲಿದ ಜಾಗಡ್ ಕಾಸ್ ತಿಕ್ಕಿಜಿ.
32	ಯಾನ್ ಧನಿಟ ಯಾನ್ ಹುಷಾರ್ ಇಜ್ಜಿ ಪಂಡ್ ಪಂಡಿನ ವಾರಡೆ ಆರ್ ಎನನ್ ಬೇಲೆಡ್ ದತ್ತೇರ್.
33	ಎನ್ನ ಧನಿ ಎನನ್ ಸುರಕ್ಷ ದೀಪಿನ ಸಾಮಾನ್ ಕೊರ್ಪೆರ್ ಪಂದ್ ಪಂಡಿ ಬೊಕ್ಕ ಅವೆನ್ ಕೊರಂದೇ ಎನಡ ಬೇಲೆ ಮಲ್ತಾವೊಂದುಲೇರ್.
34	ಎಂಕ್ ಒಂಜಿ ಪ್ರೊಮೋಷನ್ ಕೊರ್ಪೆರ್ ಪಂಡ್ ಪಾತೇರ ಕೊರ್ಯೇರ್, ಮಾತ್ರ ದಾಲ ಕಾರಣ ಕೊರಂದೇ ಬೇತೆ ಜನಕ್ ಕೊರ್ಯೇರ್.
35	ಯಾನ್ ಬೇಲಿನ್ ಬುಡಿ ಬೊಕ್ಕ ಎನ್ನ ಧನಿ ಎನ್ನ ಅಖೇರಿದ ಸಂಬಳನ್ ಕೊರೊಂದಿಜ್ಜೇರ್.
36	ಎನ್ನ ನೆರೆಕರೆದಕುಲು ಅಕ್ಕೇನ ಬೇಲಿನ್ ಎನ್ನ ಜಾಗದ ಉಲ್ಲಯಿ ಮುಟ್ಟ ಕಟ್ಟಾದೇರ್.
37	ಎನ್ನ ಕೇಣಂದೇ ಯೇರಾ ಎನ್ನ ಕುಟುಂಬದ ಜಾಗಡ್ ಬೆನ್ನಿ ದಟ್ಟೊಂದುಲೇರ್.
38	ಎನ್ನ ಜಾಗದ ಬೇಲಿನ್ ಯೇರಾ ದೆತ್ ಬುಡ್ತೇರ್.
39	ಎನ್ನ ಪಲಯೆ ಎನ್ನ ಕುಟುಂಬದ ಜಾಗದ ಒಂಜಿ ಭಾಗನ್ ಆಯೇನ ಮಾತ್ರ ಪಂಡ್ ಪಣೊಂದುಲ್ಲೆ.
40	ಎನ್ನ ಕೇಣಂದೇ ಯೇರಾ ಎನ್ನ ಜಾಗದ ಮರನ್ ಕಡ್ತೇರ್.

continued on next page

Table 4 – continued

ID	Kannada
41	ಎಂಕುಲು ನಡೆತೊಂದುತ್ತಿನಪಗೆ ಎನ್ನ ಬರಿತ್‌ಅಕ್ಲೆನ ನಾಯಿ ಎನ್ನ ಬಾಲೆನ್ ಚೂಚ್ಚಿಂಡ್.
42	ಏರೊ ಎಂಕ್ಲೆನ ಉಂತೋಂದು ಇತ್ತಿನ ಕಾರ್ಗ್ ಆಕ್ಡ್ ಪೋಯಿರ್.
43	ಒಂಜಿ ಬೇಲಿದಾಯಿ ಎನ್ನ ಮಿತ್ತ್ ಸಾಮಾನ್ ಬೂಡ್ಪಾದ್ ಎಂಕ್ ತಾಗ್ಂಡ್.
44	ಹಳ್ಳಿಡ್ ಯೇರಾ ಎನ್ನ ಬಾಗ್ಗೆ ಸುಳ್ಳು ಪಂಡ್ ಎನ್ನ ಪುದರ್ನ್ ಹಾಳ್ ಮಲ್ತೆರ್.
45	ಎನ್ನ ನೆರೆಕರೆದ ಅಕ್ಲೆನ ಪೆತ್ತಲು ಎನ್ನ ಪೂರಾ ಬುಳೆನ್ ತಿಂದ್ಡ್ ಹಾಳ್ ಮಲ್ತಂಡ್.
46	ಎಂಕ್ ನ್ಯಾಯಾಲಯದ ಒಂಜಿ ಕಾಕಜಿ ಬೈದಂಡ್, ಮಾತ್ರ ಏಟ್ ದಾದ ಪಣೊಂದುಲ್ಲೆರ್ ಪಂಡ್ ಗೊತ್ತಿಜಿ.
47	ಎಂಕ್ ನ್ಯಾಯಾಲಯಗ್ ಪೋಪಿನ ತಾರೀಖ್ ತತ್‌ಂಡ್, ದಾಯೆಗ್ ಪಂಡ ಎಂಕ್ ಏರೇಲಾ ಏಪ ಏ ಪೋಡು ಪಂಡ್ ಪಂಡಿಜೆರ್.
48	ಎಂಕ್ ಎನ್ನ ಕೆಲ್ಲದ ಧನಿನ ಮಿತ್ತ್ ಎಂಚ ದೂರು ಕೊರೋಡು ಪಂಡ್ ಗೊತ್ತಿಜಿ.
49	ಪೊಲೀಸ್-ದ ಅಕುಲು ಎನ್ನ ದೂರುನ್ ದೆತ್ತೊನೆರೆ ತಯಾರ್ ಇಜ್ಜೆರ್.
50	ಎಂಕ್ ಒಂಜಿ ವಕೀಲ್ ಬೋಡು ಪಂಡ್ ಪಂಡೆರ್, ಮಾತ್ರ ಎಂಕ್ ವಕೀಲ್ಸ್ ಎಂಚ ನಾಡೋಡು ಪಂಡ್ ಗೊತ್ತಿಜಿ.
51	ಎಂಕ್ ಗೊತ್ತಿಜಂದೇ ಯೇರಾ ಎನ್ನ ಪುದರ್ನ್ ಸಾಲ ದೆತ್ತೊನ್ನೆರ್.
52	ಎನಡ ಕೇನಂದೇ ಯೇರಾ ಎನ್ನ ಪೋಟೋನ್ ಜಾಹೀರಾತುಡ್ ಉಪಯೋಗ ಮಲ್ತೊಂದುಲ್ಲೆರ್.
53	ಯೇರಾ ಎನ್ನಲೆಕ ಆನ್‌ಲೈನ್-ಡ್ ನಟನೆ ಮಲ್ತ್ ಎನ್ನ ಕುಟುಂಬದಕ್ಲೆನ್ ಡೋಂಗಿ ಮಲ್ತೊಂದುಲ್ಲೆರ್.
54	ಪೊವಾ ಒಂಜಿ ಅಂಗಡಿ ಎನ್ನ ಕುಟುಂಬದ ಅಂಗಡಿದ ಪುದರನ್ ಉಪಯೋಗ ಮಲ್ತೊಂದುಂಡು
55	ಎಂಕ್ಲೆನ ಕುಟುಂಬ ಮಲ್ತ್ ಮಾರುನ ಸಾಮಾಂದ ಡಿಸೈನ್-ನ್ ಯೇರಾ ಕಂಡ್ಲೆರ್.
56	ಎನ್ನ ಪೋಪ ಒಂಜಿ ವಿಲ್ ಬುಡಂದೇ ತೀರ್ ಪೋಯಿ ಬೊಕ್ಕ ಎನ್ನ ಮೆಗ್ಗೆ-ಪಲಯೆನಕುಲು ಎನನ್ ಬುಡ್ಡು ಆಸ್ತಿನ ಪೂರಾ ಪಟ್ಟೊಂದುಲ್ಲೆರ್.
57	ಎನ್ನ ಅಮ್ಮನ ಬಂಗಾರ್ ಎಂಕ್ ಕೊರ್ಪೆರ್ ಪಂಡ್ ಪಂಡಿ ಬೊಕ್ಕ ಅಮ್ಮ ತೀರ್ ಪೋಯಿ ಬೊಕ್ಕ ಎನ್ನ ಪಲಯಿ ಅವೆನ್ ಪೂರಾ ದೆತ್ತೊನ್ನೆ.
58	ಎನ್ನ ಪೋಪ ತೀರ್ ಪೋಯಿ ಬೊಕ್ಕ ಎನ್ನ ಒಪ್ಪಿಗೆ ಇಜ್ಜಂದೆ ಎನ್ನ ಪೋಪ-ಅಮ್ಮನ ಇಲ್ಲನ್ ಮಾರ್ಯೆರ್ ಪಂಡ್ ಗೊತ್ತಾಂಡ್.
59	ಎನ್ನ ಮಾಮು, ಅಜ್ಜೆರ್-ನ ಜಾಗನ್ ಪಟ್ಟೆರೆ ಕೇನೊಂದಿಜ್ಜೆರ್.
60	ಎನ್ನ ಕಂಡನಿ ತೀರ್ ಪೋಯಿ ಬೊಕ್ಕ ಎನ್ನ ಕಂಡನಿನ ಕುಟುಂಬದಕುಲು ಎನನ್ ಎಂಕ್ಲೆನ ಇಲ್ಲಡ್ ಕುಲ್ಕೆರೆ ಬುಡೊಂದಿಜ್ಜೆರ್.

A.2 Priming Glossary

The priming glossary (Table 5 and Table 6) lists the vocabulary items used to construct the sparse and extensive priming blocks (§3.3). **Safe-tier** items are grammatical/functional vocabulary with no category-revealing content; **leaky-tier** items are content words (e.g. “landlord,” “money,” “land”) that overlap with terms likely to appear in specific LIST categories and are the vocabulary most responsible for the jump in accuracy under extensive priming (§5.1).

Table 5: Safe-tier glossary items (grammatical/function vocabulary, no category-revealing content).

Kannada	Romanized	Meaning
ಆಂಜೊ	aanjo	man
ಆಲ್	al	she
ಬಾಲೆ	baale	child
ಮೀನ್	meen	fish
ಪೋಪಿನಿ	popini	to go
ಮಲ್ಪುನಿ	malpuni	to do/make
ಕೊರ್ಪುನಿ	korpuni	to give
ಗೊತ್ತು	gothu	to know

Table 5 – continued

Kannada	Romanized	Meaning
ಬೋಡು	boddu	to want
ವಾಪಸ್	wapas	return/back
ಮುಗ್ಗಾವುನಿ	mugpavuni	to end/finish
ಇಜ್ಜಿ	ijji	negation(not)
ಅತ್ತ್	att	negative
ಬೊಕ್ಕ	boca	after
ದುಂಬು	dumbu	before
ಮಾತ್ರ	matra	but/contrast
ಪಂಡ	panda	quotative(THAT)
ಉಂಡು	undu	exists/is
ಅಮ್ಮ	amma	mother

Table 6: Leaky-tier glossary items (content vocabulary overlapping with LIST category terms).

Kannada	Romanized	Meaning
ಯಜಮಾನ	yejamaana	landlord/owner
ಧನಿ	dhani	landlord/employer (authority figure over you)
ಕಾಸು	kas	money
ಜಾಗ	jagga	land
ಇಲ್	ille	house
ಬೇಲಿ	belee	fence
ನೆರೆಕರೆ	nerakare	neighbor
ಪುದರ್	pudar	name
ಕಾಕಜಿ	kakaji	document
ಕುಟುಂಬ	kotumba	family
ಸಾಲ	sala	loan/debt
ಓಪ್ಪಿಗೆ	oppigay	permission

B Prompt Template

Every model call in this study, across all 19 conditions and all three models, was assembled by a single unified prompt-builder function shared across the whole pipeline (build_unified_prompt), so that script, priming tier, and RAG inclusion could vary independently without a separate, drifting prompt implementation for each condition. Each call is fully stateless: no conversation history or KV-cache continuity carries over between queries (Section 5.4). The full prompt sent to the model is assembled from up to five parts, always in this order:

1. A **static prefix**, identical across all 19 conditions: system role, LIST category definitions, an anti-collapse instruction, and two few-shot calibration examples.
2. An optional **priming block** (sparse- or full-tier conditions only), giving a script-specific Tulu vocabulary guide.
3. The **situation block**: the query itself, with a script-specific framing sentence.
4. For RAG conditions only, up to three **retrieved passages** from the Kannada court corpus.
5. **Output instructions**: a three-step reasoning scaffold ending in a top-3 JSON prediction.

B.1 Static Prefix (all 19 conditions)

You are a legal AI assistant helping classify everyday legal situations described by lay speakers into legal issue categories.

The legal categories are:

- Housing: landlord-tenant disputes, security deposits not returned, trespass into a home or dwelling, damage to rented property, eviction, someone entering your house without permission. Examples: 'My landlord did not give back my deposit', 'Someone came into my house without asking'.
- Consumer/Money/Debt: money owed but not paid, loans not repaid, overcharging beyond what was agreed, contract breaches involving payment, fraud where money was taken and the person disappeared. Examples: 'I lent money and they won't return it', 'They charged more than we agreed', 'I gave someone money and they disappeared', 'They ended our agreement early'.

- Work/Employment: wages not paid, wrongful dismissal without reason, employer stopping you from starting your own business, non-compete situations. Examples: 'My employer did not pay me this month', 'My boss told me to leave without any reason', 'My old employer is stopping me from starting my own work'.
 - Land/Property: neighbor taking your land, someone occupying your land without permission and refusing to leave, boundary disputes, land stolen after inheritance. Examples: 'My neighbor took my land', 'Someone is living on my land without asking and won't leave'.
 - Torts/Individuals: personal injury caused by another person, property damage caused by another person's negligence, someone breaking your property and not fixing it, a neighbor's tree falling on your house. Examples: 'Someone hit me with a stick and my hand hurts', 'Someone broke my fence and did not fix it', 'My neighbor's tree fell on my house'.
 - Courts/Legal-System: confusion about which court to go to, not knowing where to file a complaint, the other party being in a different location or jurisdiction, signing a document without understanding what it said. Examples: 'I don't know which court I should go to', 'The person who owes me money is in a different place', 'I signed a paper but did not understand what it said'.
 - IP/Identity: someone using your name without permission, someone selling things under your name or brand, misuse of personal or business identity. Examples: 'Someone is selling their things using my name'.
 - Estates/Wills: inheritance disputes after a family member's death, family members taking property that was meant to be given to you, land or assets disputed after someone passes away. Examples: 'After my father passed away my family took the land that was meant for me'.
- IMPORTANT: These categories describe EVERYDAY PROBLEMS, not commercial contract clauses. Classify based on what the PERSON'S SITUATION is about, NOT based on legal terminology in retrieved passages.
- Do not default to the same category repeatedly. Weigh each situation against ALL categories before deciding.
- Here are two example classifications to calibrate your reasoning:
- Example 1: Situation: "My landlord kept my deposit money after

1053	I moved out." Classification	words you recognize in the query	1112
1054	reasoning: The person had a	to understand the situation.	1113
1055	rental relationship (Housing) and	-- closing instruction, without	1114
1056	the landlord failed to return	priming:	1115
1057	money (Consumer/Money/Debt). The	No translation available. Use any	1116
1058	primary issue is the landlord-tenant	patterns you can detect in the text.	1117
1059	relationship and deposit -- Housing.		
1060	{"category": "Housing", "confidence":		
1061	0.85, "reasoning": "Landlord-tenant		
1062	deposit dispute"}]		
1063	Example 2: Situation: "Someone		
1064	broke my fence and refused to fix		
1065	it." Classification reasoning:		
1066	This is damage caused by another		
1067	individual to my property. It is		
1068	not a housing/landlord issue (no		
1069	tenancy). It is not a debt (no		
1070	money owed from an agreement). It		
1071	is physical harm/damage caused by		
1072	another person -- Torts/Individuals.		
1073	{"category": "Torts/Individuals",		
1074	"confidence": 0.80, "reasoning":		
1075	"Property damage caused by another		
1076	person's action"}]		
1077	B.2 Priming Block (sparse/full-tier		
1078	conditions only)		
1079	For conditions with a priming tier, this block		
1080	is appended immediately after the static prefix.		
1081	Sparse tier draws only from the "safe" (grammat-		
1082	ical function vocabulary) items in the glossary		
1083	(Appendix A.2); full tier draws from the entire		
1084	glossary, safe and leaky tiers combined. The		
1085	template (script-agnostic: {SCRIPT}) and each		
1086	word=meaning pair change per script and tier):		
1087	Vocabulary guide for Tulu in {SCRIPT}		
1088	script:		
1089	Nouns: {word=meaning, word=meaning,		
1090	...}		
1091	Legal nouns: {word=meaning, ...}		
1092	(full tier only)		
1093	Verbs: {word=meaning, ...}		
1094	Grammar: {word=meaning, ...}		
1095	SOV word order -- verb always at end.		
1096	Suffix -g/-k = 'to/towards'. Suffix		
1097	-n/-d = 'of/belonging to'. Quotative		
1098	marks embedded speech.		
1099	B.3 Situation Block		
1100	The framing sentence changes by script; the clos-		
1101	ing instruction changes by whether a priming block		
1102	was included.		
1103	English: SITUATION (in English):		
1104	"{display_text}"		
1105	Romanized: SITUATION (Tulu,		
1106	romanized): "{display_text}"		
1107	Kannada: SITUATION (Tulu written in		
1108	Kannada script): "{display_text}"		
1109	-- closing instruction, with priming:		
1110	No translation available. Use the		
1111	vocabulary guide above and any		
		words you recognize in the query	1112
		to understand the situation.	1113
		-- closing instruction, without	1114
		priming:	1115
		No translation available. Use any	1116
		patterns you can detect in the text.	1117
	B.4 Retrieved-Passage Block (RAG		1118
	conditions only)		1119
	The passages below are BACKGROUND		1120
	LAW ONLY. Do NOT classify what the		1121
	passage is about -- classify what the		1122
	PERSON'S SITUATION is about.		1123
	Retrieved passages from the Kannada		1124
	court corpus:		1125
	Passage 1 (similarity: {score},		1126
	source: {case_name}, LIST categories		1127
	for this case: {labels}):		1128
	{passage text}		1129
	Passage 2 ...		1130
	Passage 3 ...		1131
	Passage text and scores shown here are template		1132
	placeholders, not a fabricated example; see the		1133
	methodological note in Appendix C regarding		1134
	retrieved-passage preservation for this batch.		1135
	B.5 Output Instructions (all conditions)		1136
	Step 1: Re-read the SITUATION above.		1137
	In one sentence, describe in plain		1138
	English what problem this person is		1139
	facing -- ignore the passages for		1140
	now.		1141
	Step 2: Based on YOUR description of		1142
	the situation (not the passages),		1143
	identify the best-fitting LIST		1144
	category and explain why. Rule		1145
	out at least two other categories		1146
	explicitly.		1147
	Step 3: Provide your TOP 3 category		1148
	predictions in order, as JSON:		1149
	```json		1150
	{		1151
	"top1": {		1152
	"category": "...",		1153
	"confidence": 0.0-1.0,		1154
	"reasoning": "..."		1155
	},		1156
	"top2": {		1157
	"category": "...",		1158
	"confidence": 0.0-1.0,		1159
	"reasoning": "..."		1160
	},		1161
	"top3": {		1162
	"category": "...",		1163
	"confidence": 0.0-1.0,		1164
	"reasoning": "..."		1165
	}		1166
	}		1167
	```		1168
	CRITICAL: Every category value MUST		1169
	be exactly one of:		1170
	- Housing		1171

1172	- Consumer/Money/Debt	1233
1173	- Work/Employment	1234
1174	- Land/Property	1235
1175	- Torts/Individuals	1236
1176	- Courts/Legal-System	1237
1177	- IP/Identity	1238
1178	- Estates/Wills	1239
1179	Do NOT invent new category names or	1240
1180	use terms from the passages.	

B.6 Complete Worked Example

To ground the template above in an actual assembled call rather than placeholders, this is the complete, real prompt for Condition E_full (Kannada script, full priming, no RAG) on query 1 (human chosen label: Housing) – static prefix (Appendix B.1) followed by the full-tier Kannada priming block (built from the real glossary in Appendix A.2) and situation block, exactly as sent to the model:

1191	[...static prefix as in	
1192	Appendix B.1...]	
1193	Vocabulary guide for Tulu in Kannada	
1194	script:	
1195	Nouns: ಅಂಜೊ=man, ಆಲ್=she,	
1196	ಬಾಲೆ=child, ಮೀನ್=fish, ಅಮ್ಮ=mother	
1197	Legal nouns:	
1198	ಯಜಮಾನ=landlord/owner,	
1199	ಧನಿ=landlord/employer (authority	
1200	figure over you), ಕಾಸು=money,	
1201	ಜಾಗ=land, ಇಲ್ಲ=house, ಬೇಲಿ=fence,	
1202	ನೆರೆಕರೆ=neighbor, ಪುದರ್=name,	
1203	ಕಾಕಜಿ=document, ಕುಟುಂಬ=family,	
1204	ಸಾಲ=loan/debt, ಓಪ್ಪಿಗೆ=permission,	
1205	ದಸ್ತತ್=signature	
1206	Verbs: ಪೋಪಿನಿ=to go, ಮಲ್ಪುನಿ=to	
1207	do/make, ಕೊಪ್ಪುನಿ=to give,	
1208	ಗೊತ್ತು=to know, ಬೋಡು=to want,	
1209	ವಾಪಸ್=return/back, ಮುಗ್ಗುವುನಿ=to	
1210	end/finish	
1211	Grammar: ಇಜ್ಜಿ=negation(not),	
1212	ಅತ್ತ್=negative, ಬೊಕ್ಕ=after,	
1213	ದುಂಬು=before, ಮಾತ್ತ್=but/contrast,	
1214	ಪಂಡ=quotative(THAT),	
1215	ಉಂಡು=exists/is	
1216	SOV word order -- verb always at end.	
1217	Suffix -g/-k = `to/towards'. Suffix	
1218	-n/-d = `of/belonging to'. Quotative	
1219	marks embedded speech.	

1220	--- CLASSIFY THIS SITUATION ---	
1221	SITUATION (Tulu written in Kannada	
1222	script): "ಎನ್ನ ಧನಿ ಕಾಸ್ ವಾಪಸ್	
1223	ಕೊರ್ತೀರ್."	
1224	No translation available. Use the	
1225	vocabulary guide above and any	
1226	words you recognize in the query	
1227	to understand the situation.	
1228	No retrieved passages are provided	
1229	for this classification.	
1230	Classify based solely on the	
1231	situation description and the	
1232	category definitions above.	

REMINDER -- classify this situation:	1233
SITUATION (Tulu written in Kannada	1234
script): "ಎನ್ನ ಧನಿ ಕಾಸ್ ವಾಪಸ್	1235
ಕೊರ್ತೀರ್."	1236
No translation available. Use the	1237
vocabulary guide above and any	1238
words you recognize in the query	1239
to understand the situation.	1240
[...output instructions as in	1241
Appendix B.5...]	1242

C Representative RAG Reasoning-Trace Examples

C.1 Fact Substitution

Model: Llama3	Condition: B (Ro-	1246
	manized Tulu, no priming, RAG)	1247
Query (gloss): "Someone cut down		1248
trees on my property without permis-		1249
sion"		1250
Human chosen label: Land/Property		1251
Predicted: Consumer/Money/Debt		1252
Reasoning trace: "The situation ap-		1253
pears to be related to a financial issue,		1254
specifically a debt or payment problem,		1255
given the mention of insufficient funds		1256
and a check being returned."		1257
Verification: The third retrieved pas-		1258
sage was drawn from <i>Sridhar P. v.</i>		1259
<i>Ramesh</i> , a cheque-bounce dispute con-		1260
taining Section 138 Negotiable Instru-		1261
ments Act language and explicit ref-		1262
erences to a returned cheque (ಚೆಕ್).		1263
Neither cheques nor money are men-		1264
tioned anywhere in the query, which con-		1265
cerns unauthorized tree-felling on the		1266
speaker's land; the retrieved passage's		1267
content was substituted directly into the		1268
reasoning in place of the query's actual		1269
content.		1270
Model: Sarvam	Condition: C-sparse	1271
(Romanized Tulu, sparse priming, RAG)		1272
Query (gloss): "My roommate moved		1273
out and left me to pay all the rent"		1274
Human chosen label: Housing		1275
Predicted: Courts/Legal-System		1276
Reasoning trace: "The person is ac-		1277
tively involved in a legal process, men-		1278
tioning filing a document, producing ev-		1279
idence (WhatsApp screenshots), and un-		1280
dergoing a court procedure (fingerprint-		1281
ing)."		1282
Verification: The first retrieved passage		1283

(*Amith Ramachandra v. Shantha*) contains both details verbatim: ವಾಟ್ಸಾಪ್ ಸಂದೇಶಗಳ ಸ್ಕ್ರೀನ್ ಶಾಟ್ (WhatsApp message screenshots), and (typing, which the model misunderstood as fingerprinting). Nothing resembling either detail appears in the query (a roommate/rent dispute).

C.2 Confabulation

Model: Llama3 **Condition:** D-full (Kannada script, full priming, RAG)
Query (gloss): “The people next door make so much noise I cannot sleep at night”
Human chosen label: Housing
Predicted: Courts/Legal-System
Reasoning trace: “The situation involves threats and demands, which suggests that legal action may be involved or necessary to resolve the issue.”
Verification: Neither “threats” nor “demands” (nor Kannada equivalents ಬೆದರಿಕೆ / ಬೇಡಿಕೆ) appear in the query or in any of the three retrieved passages (*Amith Ramachandra v. Shantha*; *Sadashiva Shetty v. Vijaya Bank*; *State v. Sudhakar Pai*) (though threats appear in the *Amith Ramachandra v. Shantha* text in a separate chunk). The reasoning invents a threat scenario with no basis anywhere in the available context.

Model: Llama3 **Condition:** F (Kannada script, no priming, RAG)
Query (gloss): “My neighbor’s dog bit my child while we were walking”
Human chosen label: Torts/Individuals
Predicted: Estates/Wills
Reasoning trace: “The situation involves eviction and family members, which suggests an issue related to property or inheritance.”
Verification: No eviction-related term (ಖಾಲಿ, ಉಚ್ಚಾಟನೆ) appears in the query or in any of the three retrieved passages (*Nikitha A. v. Santhosh Ghatke*; *Akkamma v. Ramesh Patil*; *Dr. Alfred Aranha v. Sahel Peter Aranha*). “Eviction” is introduced with no basis in either source, though family members are brought up in both *Nikitha A. v. Santhosh Ghatke* and *Akkamma v. Ramesh Patil*.

C.3 RAG Content and Correct Classification Aligning

Model: Sarvam **Condition:** F (Kannada script, no priming, RAG)
Query (gloss): “My neighbor’s dog bit my child while we were walking”
Human chosen label: Torts/Individuals
Predicted: Torts/Individuals
Reasoning trace: “This is a personal injury case caused by another person’s (or their animal’s) negligence or action, which falls under the Torts/Individuals category.”
Verification: The retrieved passage (*Nikitha A. v. Santhosh Ghatke*, similarity 0.916) is actually on track with Torts/Individuals content. Llama3 confabulates on this same query under the same condition (see above), predicting Estates/Wills with reasoning citing “eviction and family members”. The relevance of a retrieved passage does not ensure accurate classification.

Model: Hex-1 **Condition:** A (English gloss, RAG)
Query (gloss): “After my father passed away my family took the land that was meant to be given to me”
Human chosen label: Estates/Wills
Predicted: Estates/Wills
Reasoning trace: “The situation involves inheritance disputes after a family member’s death.”
Verification: The retrieved passage (*Gagan B.S. v. Shivamurthy T.*, similarity 0.824) is itself an ancestral-property inheritance-share dispute (ಪಿತೃಜ್ಞಾಪ್ತಿ, “ancestral property”). Retrieval and classification align correctly here as well.

C.4 Relevant Retrieval, Persistent Failure

Model: Sarvam **Condition:** D-full (Kannada script, full priming, RAG)
Query (gloss): “They ended our agreement before the time we decided”
Human chosen label: Consumer/Money/Debt
Predicted: Land/Property
Reasoning trace (translated): “This situation involves a contract dispute

related to property rights or control. This is more of a property and contract dispute than a landlord-tenant relationship.”

Verification: The retrieved passage (*Sridhar P. v. Ramesh*, similarity 0.911) is an on-topic financial/debt dispute, though financial matters are discussed briefly. Despite having retrieved directly relevant material, the model still produced an unrelated category.

D Full Condition-Level Results

This appendix gives the complete comprehension and RAG results underlying the condensed Findings in §4: the full script $\times$ priming-tier breakdown (Table 7, Figure 3), the full 19-condition breakdown (Table 8, Figure 4), and the complete significance test battery (Table 9).

Script	Tier	Llama3		Hex-1		Sarvam	
		Hits@1	Hits@3	Hits@1	Hits@3	Hits@1	Hits@3
Romanized	bare	16.7%	46.7%	16.7%	65.0%	33.3%	61.7%
	sparse	13.3%	43.3%	18.3%	55.0%	30.0%	60.0%
	full	31.7%	56.7%	25.0%	61.7%	40.0%	71.7%
Kannada	bare	46.7%	76.7%	41.7%	68.3%	48.3%	76.7%
	sparse	51.7%	75.0%	45.0%	76.7%	46.7%	78.3%
	full	60.0%	83.3%	60.0%	80.0%	66.7%	88.3%
Tamil	bare	16.7%	48.3%	28.3%	66.7%	28.3%	55.0%
	sparse	15.0%	51.7%	33.3%	61.7%	33.3%	56.7%
	full	33.3%	73.3%	35.0%	65.0%	35.0%	68.3%
Malayalam	bare	21.7%	53.3%	25.0%	60.0%	21.7%	51.7%
	sparse	20.0%	55.0%	28.3%	68.3%	38.3%	68.3%
	full	40.0%	76.7%	48.3%	73.3%	51.7%	76.7%

Table 7: Comprehension accuracy by script and priming tier, no RAG, all three models ($n = 60/\text{cell}$).

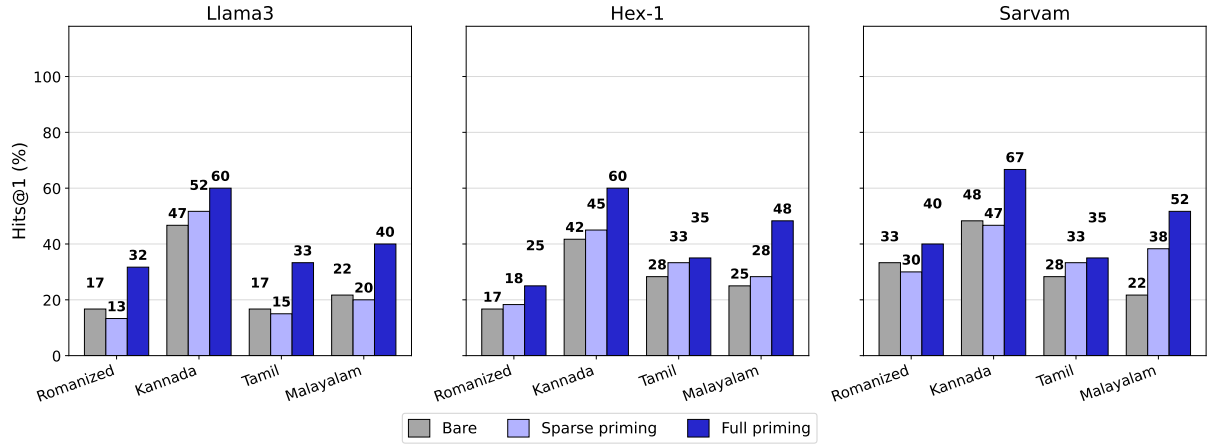


Figure 3: Hits@1 by script and priming tier, per model ($n = 60/\text{bar}$).

Condition	Llama3 H1	Llama3 H3	Hex-1 H1	Hex-1 H3	Sarvam H1	Sarvam H3	n
A	61.7%	95.0%	86.7%	98.3%	91.7%	100.0%	60
B	16.7%	56.7%	16.7%	50.0%	28.3%	63.3%	60
C_sparse	16.7%	46.7%	11.7%	46.7%	31.7%	61.7%	60
C_full	18.3%	53.3%	25.0%	63.3%	45.0%	73.3%	60
D_sparse	26.7%	66.7%	41.7%	68.3%	55.0%	75.0%	60
D_full	23.3%	68.3%	46.7%	80.0%	60.0%	80.0%	60
E_sparse	51.7%	75.0%	45.0%	76.7%	46.7%	78.3%	60
E_full	60.0%	83.3%	60.0%	80.0%	66.7%	88.3%	60
F	25.0%	60.0%	38.3%	71.7%	53.3%	71.7%	60
G	16.7%	46.7%	16.7%	65.0%	33.3%	61.7%	60
H	46.7%	76.7%	41.7%	68.3%	48.3%	76.7%	60
I	16.7%	48.3%	28.3%	66.7%	28.3%	55.0%	60
J	21.7%	53.3%	25.0%	60.0%	21.7%	51.7%	60
K_sparse	15.0%	51.7%	33.3%	61.7%	33.3%	56.7%	60
K_full	33.3%	73.3%	35.0%	65.0%	35.0%	68.3%	60
L_sparse	20.0%	55.0%	28.3%	68.3%	38.3%	68.3%	60
L_full	40.0%	76.7%	48.3%	73.3%	51.7%	76.7%	60
M_sparse	13.3%	43.3%	18.3%	55.0%	30.0%	60.0%	60
M_full	31.7%	56.7%	25.0%	61.7%	40.0%	71.7%	60

Table 8: Hits@1 / Hits@3 across all 19 conditions, per model ($n = 60$ each).

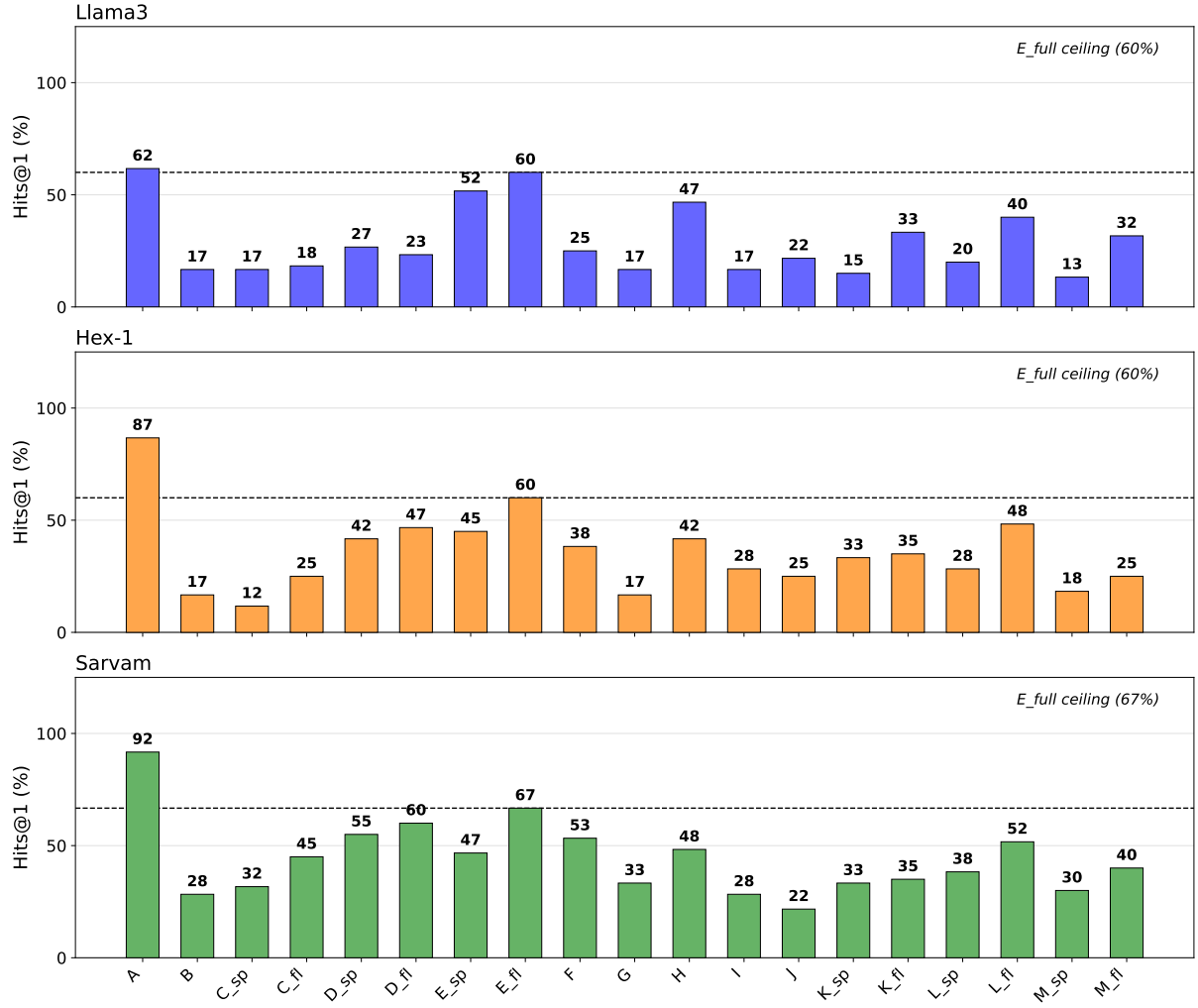


Figure 4: Hits@1 across all 19 conditions, per model ($n = 60/\text{bar}$). Each panel’s dashed line marks that model’s own no-RAG, full-priming ceiling (E_{full}).

Comparison (Hits@1)	Llama3	Hex-1	Sarvam
<i>Baseline comprehension (binomial test vs. chance/majority)</i>			
H (Kannada bare) vs. chance (12.5%) / majority (25%)	$p < .001 / p < .001$	$p < .001 / p < .001$	$p < .001 / p < .001$
D_full vs. majority-class (25%)	$p = .665$ (ns)	$p < .001^{***}$	$p < .001^{***}$
<i>Priming effects (Fisher’s exact)</i>			
E_full vs. H (Kannada full-priming effect)	$p = .200$ (ns)	$p = .067$ (ns)	$p = .064$ (ns)
L_full vs. J (Malayalam full-priming effect)	$p = .047^*$	$p = .013^*$	$p = .001^{**}$
<i>RAG effects (Fisher’s exact)</i>			
H vs. F (RAG collapse, Kannada bare)	$p = .022^*$	$p = .852$ (ns)	$p = .715$ (ns)
E_sparse vs. D_sparse (RAG collapse, Kannada sparse)	$p = .009^{**}$	$p = .854$ (ns)	$p = .465$ (ns)
E_full vs. D_full (RAG collapse, Kannada full)	$p < .001^{***}$	$p = .200$ (ns)	$p = .570$ (ns)
A vs. D_full (query language under RAG)	$p < .001^{***}$	$p < .001^{***}$	$p < .001^{***}$

Table 9: Fisher’s exact and binomial tests for pairwise condition comparisons at Hits@1, $n = 60/\text{condition}$. Grouped by comparison type: baseline comprehension checks against chance/majority guessing, priming-effect checks (no RAG), and RAG-effect checks (Kannada bare, sparse, and full priming, plus query-language effect under RAG). $^*p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Model	Bare	Sparse prime	Full prime	RAG only	RAG+sparse	RAG+full
Llama3	46.7>16.7***	51.7>13.3***	60.0>31.7**	25.0>16.7 (ns)	26.7>16.7 (ns)	23.3>18.3 (ns)
Hex-1	41.7>16.7**	45.0>18.3**	60.0>25.0***	38.3>16.7*	41.7>11.7***	46.7>25.0*
Sarvam	48.3>33.3 (ns)	46.7>30.0 (ns)	66.7>40.0**	53.3>28.3**	55.0>31.7*	60.0>45.0 (ns)

Table 10: Kannada-script vs. Romanized-script comprehension (Hits@1, %) across all six priming/RAG tiers, per model ($n = 60$ /condition). Values shown as Kannada% > Romanized%. Fisher’s exact test, same methodology as Table 9. Computed from condition-level results in Table 8; no new model calls were made. * $p < .05$, ** $p < .01$, *** $p < .001$.

Model	Tier	E-right/D-wrong	D-right/E-wrong	Ratio
Llama3	full	26	4	6.5:1
Llama3	sparse	20	5	4.0:1
Hex-1	full	10	2	5.0:1
Hex-1	sparse	8	6	1.3:1
Sarvam	full	7	3	2.3:1
Sarvam	sparse	4	9	0.4:1

Table 11: Disagreement-level analysis: among cases where full/sparse-priming no-RAG (E) and RAG (D) predictions diverge, how often each condition was uniquely correct, per model and priming tier ($n = 60/\text{condition}$).

Judge language	Hits@1	Hits@3	Passages retained	No-RAG fallback
Kannada script	25.0% (5/20)	65.0% (13/20)	2.85 / 3	0/20
English gloss	60.0% (12/20)	80.0% (16/20)	1.65 / 3	3/20

Table 12: CRAG relevance-verification results, Llama3, Kannada-script D-condition queries ($n = 20$). Fisher’s exact test on Hits@1 (English-judge vs. Kannada-judge): $p = .054$. Reference points from the full 60-query set (Table 8): D_full (RAG, full priming) = 23.3%; E_full (no-RAG, full priming) = 60.0%.

E Court Cases Cited

Table 13 lists full identifiers for the eight court cases named in Appendix C, drawn from the 18-document Kannada legal corpus (§3.2). All documents are public lower-court filings retrieved from ecourtsindia.com; CNR is the eCourts Case Number Record, the portal’s unique case identifier.

Case	Case No.	Court	CNR / Date
Sridhar P. v. Ramesh	C.C. No. 5343/2024	4th Addl. Small Causes Court & ACJM, Bengaluru Urban	KABC020187742024, 3 Jun 2025
Amith Ramachandra v. Shantha	P.C.R. No. 3981/2025	41st Addl. Chief Judicial Magistrate, Bengaluru Urban	KABC030166202025, 6 May 2025
Sadashiva Shetty v. Vijaya Bank	R.A. No. 43-2001	Principal Civil Judge (Sr. Div.) & CJM, Udupi	KAUP020002942001, 24 Jan 2009
State v. Sudhakar Pai	C.C. No. 1357/2016	JMFC 2nd Court, Mangaluru	KADK070014602016, 18 Oct 2016
Nikitha A. v. Santhosh Ghatke	CrI.Misc. 127/2023	JMFC Traffic Court VI, Bengaluru Urban	KABC080042862023, 4 Jul 2025
Akkamma v. Ramesh Patil	CrI.Misc. 17/2018	Principal Civil Judge & JMFC, Byadgi, Haveri	KAHV210000762018, 1 Aug 2018
Dr. Alfred Aranha v. Sahel Peter Aranha	O.S. No. 419/2013	4th Addl. Senior Civil Judge & 1st Class JMFC, Mangaluru	KADK030000092013, 2 Feb 2019
Gagan B.S. v. Shivamurthy T.	Property partition suit	Senior Civil Judge & JMFC, Shikaripura, Shivamogga	KASM510014042020, 16 Aug 2025

Table 13: Court cases named in Appendix C, with case number, deciding court, eCourts CNR identifier, and order date. All are public lower-court filings from ecourtsindia.com.